THE SMOOTHED STAPLE: SHORTER CERTIFIED ESCAPE PATHS FOR BELLMAN'S LOST-IN-A-FOREST PROBLEM IN ISOSCELES TRIANGLES

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ABSTRACT. For the isosceles triangle T with unit legs and base angle β, Temerev (2026) proved the certified upper bound $\ell(T) \leq 1.2849615334\ldots$ at the golden gnomon ($\beta = 36°$) via a three-segment "staple" path, together with an interval theorem for $\beta \in [31.08°, 42.16°]$. We show that, at the golden gnomon, the three-segment staple is not optimal among convex-position polygonal paths satisfying the same escape criterion. Using the same exact reduction (escape of a path follows from a single polygon-contains-disk inclusion for a weighted Minkowski sum W of rotated copies of the path's convex hull), we exhibit an explicit 20-vertex path with rational coordinates proving

$$\ell(T) \;\leq\; 1.2826879686\ldots \;<\; 1.2849615334\ldots$$

at the golden gnomon, with every accepted sign determination carried out in exact arithmetic over $\mathbb{Q}(\sqrt{5}, \sqrt{10-2\sqrt{5}})$. Numerical optimization suggests that the underlying continuum object is a symmetric *smoothed staple* whose shoulders are circular arcs, and a bang-bang argument for the associated infinite-dimensional linear program in support functions leads us to conjecture that their curvature radius is exactly $\sin\beta$. Conservative per-angle computations indicate the family beats the scaled Zalgaller caliper down to $\beta \approx 30.76°$ (staple: $31.05°$). Optimality is not claimed.

1. Introduction

Bellman's lost-in-a-forest problem asks for the shortest path guaranteed to reach the boundary of a convex region of known shape from an unknown starting point and heading [1]. For isosceles triangles with base angles below $45°$ the problem is open; Ward [6] compared the classical candidates numerically, and Temerev [5] proved the first certified improvement on an interval

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Status. This is an informal working draft, circulated for verification only. It is not submitted, and not intended for submission in its present form. Its purpose is to state precisely what the accompanying machine-checkable certificates verify, and on what imported result they depend, so that the argument can be checked by others. The reasoning behind the certificates — in particular the variant of the verification scheme described in Section 2 — has not been widely reviewed. Comments and corrections are very welcome.

of base angles via a piecewise-linear three-segment family of *staples*, including the bound $\ell(T) \le 1.2849615334\ldots$ at the golden gnomon ($\beta = 36°$, unit legs).

This note improves those bounds. The key observation is structural: the escape criterion used in [5] depends only on the convex hull of the path, so the staple is a single point in the much larger space of paths with convex-position vertices. Optimizing freely in that space, under the *same exact criterion*, produces strictly shorter escaping paths, and the limiting shape is not piecewise linear.

2. Preliminaries: the exact escape reduction

Let $T = T(\beta)$ have unit legs, base angle β, outward edge normals $n_0 = (0, -1)$, $n_{1,2} = (\pm \sin\beta, \cos\beta)$ with weights (edge lengths) $w_0 = 2\cos\beta$, $w_1 = w_2 = 1$, and let $\rho = 2\,\mathrm{Area}(T) = \sin 2\beta$. For a convex body H put

$$W(H) \;=\; \sum_{i=0}^{2} w_i\, R_{n_i} H,$$

the Minkowski sum of rotated copies of H, where R_n is the rotation taking $(1, 0)$ to n's frame as in [5]. Since $\sum_i w_i R_{n_i} = 0$, the construction is translation invariant, and since plane rotations commute, the criterion below is rotation invariant.

Proposition 1 (imported from [5], via Lutwak [4])**.** *If* $\mathrm{disk}(\rho) \subseteq W(H)$, *where H is the convex hull of a path P, then P escapes $T(\beta)$.*

All certificates below verify the inclusion of Proposition 1 for an explicit polygon, by exhibiting a convex, positively oriented cycle of points of W whose edge lines all lie at distance $\ge \rho$ from the origin. Because the cycle's vertices belong to W and W is convex, the cycle's hull is contained in W; hence the disk inclusion follows from the finitely many edge conditions $\mathrm{cross}(p, q)^2 - \rho^2\,|q - p|^2 > 0$, each an exact sign determination in $\mathbb{Q}(\sqrt{5}, \sqrt{10 - 2\sqrt{5}})$ at $\beta = 36°$.

3. Main theorem

Theorem 2. *Let T be the golden gnomon (apex $108°$, base angles $36°$, unit legs). The 20-vertex polygonal path with the rational vertices of Appendix A (traversed in order) escapes T, and consequently*

$$\ell(T) \;\le\; 1.2826879686\ldots \;<\; 1.2849615334\ldots$$

Proof (machine-checked). The escape inclusion is verified by `verification/verify_certificate20.py` in the accompanying repository: the proposed 57-vertex cycle of Minkowski points satisfies, for every edge, positive convexity cross product, positive orientation cross product, and $D = \mathrm{cross}^2 - \rho^2 |e|^2 > 0$, all by exact symbolic sign determinations (`sympy`); the minimum edge distance is $0.95105745\ldots > \rho = \sin 72° = 0.95105651\ldots$. The path length, a sum of 19 square roots of rationals, is bounded above by exact rational majorants of each square root (integer arithmetic), and the resulting rational is compared exactly against $12826879687/10^{10}$. The certificate cycle is stored explicitly with the data; floating point appears in no accepted check. □

Remark 3. A shorter, fully self-contained 8-vertex certificate proving $\ell(T) \leq 1.2827871$ is included for independent auditing (`smoothed_staple_certificate.py`).

4. The continuum object and the $\sin\beta$ curvature law

Free polyline optimization under the exact criterion yields the strictly decreasing values 1.2849609 ($n = 4$, the staple), 1.2829755 ($n = 6$), 1.2827742 ($n = 8$), 1.2827491 ($n = 10$). A piecewise line/arc model with free radii, solved by constraint generation on the support-function form of the inclusion, converges (dense margin $-6 \cdot 10^{-15}$) to the symmetric path

$$\underbrace{0.291027}_{\text{straight}} \to \underbrace{r = 0.587785,\ \varphi = 9.5232^\circ}_{\text{arc}} \to \underbrace{0.071683}_{\text{straight}} \to \underbrace{76.3054^\circ}_{\text{corner}} \to \underbrace{0.361665}_{\text{base}} \to \text{(mirror)},$$

of total length $L_\infty = 1.2826726243$. The arc radius agrees with $\sin 36^\circ = 0.5877853$ to the optimizer's precision, with the radius left free; a second arc allowed on each shoulder degenerates.

Relation to prior work. The line–arc geometry is closely related to Gibbs's "Tunnel" solutions, proposed on Monte Carlo evidence for base angles roughly 30.8°–38.1° [2] — a range containing 36° — and coupled curvature conditions for extremal escape paths appear in his analysis for convex polygons [3]. Those proposals were not accompanied by certified bounds. The contribution of the present note is therefore *not* the qualitative smoothed geometry, which should be credited to Gibbs, but the exact rational polygonal certificate showing that a smoothed/Tunnel-type path rigorously improves the previously certified staple bound at the golden gnomon, and the specific quantitative prediction $r = \sin\beta$ below.

The value $\sin\beta$ is predicted by the following argument. The constraint of Proposition 1 reads $h_W(u) = \sum_i w_i\, h_H(R_{n_i}^T u) \geq \rho$ for all directions u and is *linear* in the support function h_H; path length is likewise a linear functional of h_H over the traversed normal window. The problem is therefore an infinite-dimensional linear program. On a window where the constraint is active, $h_W \equiv \rho$ gives $h_W + h_W'' = \rho$, i.e.

$$\sum_i w_i \left(h_H + h_H''\right)(\theta - \alpha_i) \;=\; \rho,$$

so the curvature radii of ∂H at the three rotated angles combine, with weights w_i, to the constant ρ. A bang-bang optimum concentrates all curvature on the copy with the largest weight $w_0 = 2\cos\beta$, giving curvature radius

$$\frac{\rho}{2\cos\beta} \;=\; \frac{\sin 2\beta}{2\cos\beta} \;=\; \sin\beta.$$

Conjecture 4. *For β in the staple regime, the optimal escape path of $T(\beta)$ is a concatenation of line segments, corners, and circular arcs of curvature radius exactly* $\sin\beta$*, symmetric about the axis of T; at $\beta = 36^\circ$ its length is* $1.2826726\ldots$

5. Per-angle behaviour and the caliper crossing

Conservative per-angle values (grid-optimized 8-vertex polylines followed by a conservative scaling correction evaluated in floating point against the hull-edge criterion; unlike the certificates of Section 3 these corrections are numerical, not symbolic) give a uniform gain of $\approx +0.0026$ over the per-angle optimal staple for $\beta \in [30^\circ, 33^\circ]$, moving the crossing with the scaled Zalgaller caliper from $\beta \approx 31.05^\circ$ (staple) to

$$\beta \;\approx\; 30.76^\circ \qquad \text{(conservative; the continuum crossing is lower still).}$$

The family also beats the low-regime square path at every sampled angle down to 29°; square comparisons are made only below Ward's regime switch $\beta_0 \approx 39.13^\circ$. Near the upper end the smoothing gain genuinely collapses ($+1.6 \cdot 10^{-4}$ at 42.0°, $+10^{-5}$ at 42.5°; $n = 10$ polylines with exact-margin correction, the same recipe that captures $\approx 90\%$ of the $2.3 \cdot 10^{-3}$ gain at 36°): the arcs switch off as β approaches the zee crossing, which therefore stays at $\beta \approx 42.284^\circ$, essentially the staple's 42.28° and short of the conjectured 42.3°.

6. Open problems

(i) An interval theorem for the smoothed family in $u = \tan(\beta/2)$, replacing arcs by circumscribed tangent polygons and reusing the Sturm certificate machinery of [5]; (ii) the exact KKT system at the gnomon with $r = \sin\beta$ imposed (five unknowns), which may yield a closed form for $1.2826726\ldots$; (iii) a proof of Conjecture 4; (iv) matching lower bounds.

Appendix A. Certificate vertices

The 20 vertices (path order) of the Theorem 2 witness; denominators 10^8 or divisors thereof.

vertex	x	y
v_0	$-1256697/12500000$	$9697297/25000000$
v_1	$-18524027/100000000$	$11076737/100000000$
v_2	$-18800781/100000000$	$10171311/100000000$
v_3	$-600917/3125000$	$4309011/50000000$
v_4	$-4903679/25000000$	$1763883/25000000$
v_5	$-99787/500000$	$5482169/100000000$
v_6	$-20257379/100000000$	$155879/4000000$
v_7	$-20513533/100000000$	$2305271/100000000$
v_8	$-10785997/50000000$	$-5656137/100000000$
v_9	$-5394019/25000000$	$-568721/10000000$

v_{10}	12159143/100000000	−9379621/50000000
v_{11}	193587/1562500	−18428471/100000000
v_{12}	8348493/50000000	−12229211/100000000
v_{13}	17464737/100000000	−11066231/100000000
v_{14}	9100817/50000000	−4945211/50000000
v_{15}	9453909/50000000	−8702301/100000000
v_{16}	19584521/100000000	−3750221/50000000
v_{17}	5057973/25000000	−6284817/100000000
v_{18}	20847673/100000000	−2530147/50000000
v_{19}	33614857/100000000	10934787/50000000

References

[1] S. R. Finch, J. E. Wetzel, *Lost in a forest*, Amer. Math. Monthly 111 (2004), 645–654.
[2] P. E. Gibbs, *Lost in an Isosceles Triangle*, working paper (July 2016).
[3] P. E. Gibbs, *Bellman's Escape Problem for Convex Polygons*, working paper (June 2016).
[4] E. Lutwak, *Containment and circumscribing simplices*, Discrete Comput. Geom. 19 (1998), 229–235.
[5] A. Temerev, *An improved upper bound for Bellman's lost-in-a-forest problem in isosceles triangles* (2026), `https://github.com/atemerev/bellman-staple`.
[6] J. W. Ward, *Exploring the Bellman forest problem*, manuscript (2008).